The Physician and Sportsmedicine

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/ipsm20

Risk Factors for Acute Ankle Sprains in Field-Based, Team Contact Sports: a Systematic Review of Prospective Etiological Studies

Patrick Dolan, Ian Kenny, Liam Glynn, Mark Campbell, Giles D Warrington, Roisin Cahalan, Andrew Harrison, Mark Lyons & Thomas Comyns

To cite this article: Patrick Dolan, Ian Kenny, Liam Glynn, Mark Campbell, Giles D Warrington, Roisin Cahalan, Andrew Harrison, Mark Lyons & Thomas Comyns (2022): Risk Factors for Acute Ankle Sprains in Field-Based, Team Contact Sports: a Systematic Review of Prospective Etiological Studies, The Physician and Sportsmedicine, DOI: 10.1080/00913847.2022.2093618

To link to this article: https://doi.org/10.1080/00913847.2022.2093618

Accepted author version posted online: 27 Jun 2022.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ipsm20

Check for updates

Publisher: Taylor & Francis & Informa UK Limited, trading as Taylor & Francis Group

Journal: *The Physician and Sportsmedicine*

DOI: 10.1080/00913847.2022.2093618

Formatted: Font: Bold
Formatted: Font: Bold
Formatted: Font: Not Bold, Italic
Formatted: Font: Bold
Formatted: Left, Space Before: 12 pt, After: 6 pt, Line spacing: single

Risk Factors for Acute Ankle Sprains in Field-Based, Team Contact Sports: a Systematic Review of Prospective Etiological Studies

Patrick Dolan[1,2]* (Corresponding Author), Ian Kenny[1,2,7], Liam Glynn[3,8], Mark Campbell[1,4,7], Giles D Warrington[1,7], Roisin Cahalan[5,6], Andrew Harrison[7], Mark Lyons[1,7], Thomas Comyns[1,2,7]

[1] Department of Physical Education and Sport Sciences, University of Limerick, Limerick, Ireland
[2] Health Research Institute, University of Limerick, Limerick, Ireland
[3] School of Medicine, University of Limerick, Limerick, Ireland
[4] Lero, The Irish Software Research Centre, University of Limerick, Limerick, Ireland
[5] School of Allied Health, University of Limerick, Limerick, Ireland
[6] Physical Activity for Health Research Cluster, University of Limerick, Limerick, Ireland
[7] Sport and Human Performance Research Centre, University of Limerick, Limerick, Ireland
[8] HRB Primary Care Clinical Trials Network, Ireland

Patrick Dolan (*corresponding author)
Department of Physical Education and Sport Sciences, University of Limerick
Limerick, Ireland
V94-YDE9, PG-034
Email: Patrick.dolan@ul.ie
Tel.: (+1)2158964961
ORCID 0000-0002-8089-0343

ACCEPTED MANUSCRIPT

Disclaimer: This manuscript is original and is not being considered for publication elsewhere. The authors listed have contributed significantly to the given text and consented to have their names appear for authorship. There are no conflicts of interest related to this research and no external financial support. All procedures set out in this paper have been approved by the University of Limerick Research Ethics Committee.

Manuscript length : 5375 words (excluding abstract)
Abstract length : 294 words
Figures : 1
Tables in text: 5

[1] Department of Physical Education and Sport Sciences, University of Limerick, Limerick, Ireland
[2] Health Research Institute, University of Limerick, Limerick, Ireland
[3] School of Medicine, University of Limerick, Limerick, Ireland
[4] Lero, The Irish Software Research Centre, University of Limerick, Limerick, Ireland
[5] School of Allied Health, University of Limerick, Limerick, Ireland

Supplementary tables: 2

Conflicts of interest: none to declare

Risk Factors for Acute Ankle Sprains in Field-Based, Team Contact Sports: a Systematic Review

Objective: The aim of this systematic review was to identify prospectively measured ankle sprain risk factors in field-based team contact sports.
Methods: Eight databases including SPORTDiscus, CINAHL Complete, MEDLINE (EBSCO), Education Source, Web of Science, Scopus, Embase and Pubmed were searched using specific Boolean terms. A modified-CASP diagnostic test assessed the quality of the included studies. Extensive data extraction included but was not limited to injury definition, protocols for injury diagnosis and recording, and outcomes associated with ankle sprain.
Results: 4012 records were returned from the online search and seventeen studies met the inclusion criteria for this review. Twelve different risk factors including anatomic alignment of the foot and ankle, joint laxity, height, mass, BMI (body mass index), age, ankle strength, hip strength, single leg landing performance (ground reaction force, pelvic internal rotation and knee varus), and single leg reach were all found to be associated with ankle sprain incidence. Injury definitions and methods of diagnosis and recording varied across the 17 studies.
Conclusion: This review updates the literature on prospective risk factors for ankle sprain in a specific population rather than heterogeneous cohorts previously studied. From more than twenty categories of risk factors investigated for ankle sprain association across seventeen studies in field-based team contact sports, 12 variables were found to be associated with increased incidence of ankle sprain. In order to reduce the risk of ankle sprain, BMI, ankle plantar and dorsiflexion strength, hip strength, and single leg landing performance should be factored in to athlete assessment and subsequent program design. More studies utilizing standardized definitions and methods of recording and reporting are needed. Future prospective etiological studies will allow strength and conditioning coaches, physiotherapists and physicians to apply specific training principles to reduce the risk and occurrence of ankle sprain injuries.

Key Words ankle athletic injuries, athletic performance, epidemiology, etiology, risk factors, sports, sports medicine, sprains and strains

ACCEPTED MANUSCRIPT

1 INTRODUCTION

It is widely accepted today that there are multiple physiological, psychological and behavioral benefits to participating in physical activity and sport in most populations [1-11]. While the level of benefit varies across populations[12], there is inherent risk associated with participating in physical activity particularly in contact sports [13]. Injuries, particularly to the ankle, have shown an increase in incidence in the last few decades [14] and represent the most frequently injured body location for sports injuries presented to emergency departments in multiple countries [15, 16].

The nature of injuries occurring at the ankle include sprains, fractures, dislocations, and ruptures [17]. Sprains have been recorded at two million occurrences per year in the United States, where half (49.3%) are attributed to athletic activities [18]. In sporting populations specifically, ankle sprains are responsible for 15% of all diagnosed injuries [19]. This is the most common injury diagnosed (relative to location and type), in most sports, at different competitive levels, across many countries [15, 19-22]. Furthermore, a study in North American high school athletes investigating costs assessed by quality adjusted life years, discovered that across three years and 648 ankle injuries, of which sprains were most common, there was a mean comprehensive cost of $11,925 per injury [23]. By way of international comparison, data from the Netherlands revealed that, although each ankle sprain was associated with €360 ($507) for direct and indirect cost, their 234,000 incidences per year equated to an estimated €84,240,000 annual expense [24].

The ankle complex is organised by two primary sub-joints, the distal tibiofibular which is a fibrous syndesmosis joint, and the talocrural, a uniaxial synovial hinge; each with multiple supporting ligaments [25]. The most common mechanism of injury is ankle inversion and as a result the anterior talofibular ligament has shown to be the most frequently involved ligament in ankle sprains [26, 27]. However, due to the sophistication of this multi-joint complex, damage to a variety of structures can happen with differing associated diagnoses. A sprain injury is defined as structural damage to a ligament. Notably, the three types of ankle sprain are inversion or lateral ankle sprain (LAS), eversion or medial ankle sprain (MAS), and syndesmosis or high ankle sprain (HAS) [28, 29].

In addition to the established prevalence, studies investigating several sports have also shown that ligamentous injuries, compared to muscle and tendon (strains), may have the highest severity, that is days lost from participation in training or matches (time-loss) [30, 31]. A recent study investigating injury burden in inline-hockey players representing ten teams and both sexes, found that among all severe injuries (≥29days lost) [32] sprains of the ankle presented the highest prevalence (16%), with ankle fractures second most common (8%). Field-based sports also report high severities from ankle sprain. A recent investigation over

ACCEPTED MANUSCRIPT

three seasons reported 24.7 average days lost for ankle sprains in rugby union[33], where another study in soccer found an average loss of 33 days for lateral ankle sprains [22]. Although many sport settings may lack the resources for regular radiological examination of each ankle injury, a report from the 2016 Summer Olympic Games in Rio de Janeiro showed that 10% of all MRIs (magnetic resonance images) performed were at the ankle, of which 45% revealed ligamentous injuries [27]. In addition to time-loss, consequences include but are not limited to the direct and indirect costs of diagnosis, treatment and management [34], long-term physical complaints even after 5-years of onset (i.e. chronic ankle instability) [35], and risk of recurrence [36].

There has long been a strong focus on injury prevention within the sports science community, first presented in 1992 by Van Mechelen et al. and their four step process [37], and an updated model by Finch in 2006 [38]. In injury epidemiology research and practice, a primary goal is to identify risk factors that may detect early deficits or abnormalities in movement and human performance, which may lead to an increased risk of sustaining injury. These risk factors may pertain to the athlete directly (intrinsic) or their environment (extrinsic). As described by Taimela et al. [39], intrinsic factors (e.g. proprioception) are characteristics which stem from the individual themselves, and predispose oneself to an injury; whereas extrinsic risk factors (e.g. footwear) are independent of the individual but can be related to the onset of injury. Both intrinsic and extrinsic risk factors may be considered modifiable (e.g. strength) or non-modifiable (e.g. leg length) [40]. Extrinsic factors such as playing surface, weather, time zone and travel, have been previously investigated for both risk of injury [41] and player performance [42].

Ankle sprains are often reported as the most common injury among athletes, therefore it is necessary for applied practitioners such as strength and conditioning coaches and physiotherapists to know what are the risk factors for ankle injury associated with their particular sport (i.e., field-based, team contact sport [43]). While previous reviews have been conducted regarding ankle sprains, they tend to focus on treatment and management [44-47] or risk of recurrence and long-term effects [48, 49]. For reviews which have identified associated risk factors, some address only one type of ankle sprain [50-52], and others require updating with more recent literature [50, 53]. Therefore, this review aims to fill the gap regarding ankle sprain etiology and update the current literature. Rather than identifying risk factors across many types of sport or physical activity levels [50-54], similar to Ryan et al's [55] research into groin injury risk factors, this review sought to identify associated risk factors for ankle sprains within field-based, team contact sports.

2 METHODS

2.1 Literature search methodology

The Prisma-P checklist and Explanation and Elaboration was used for reporting [56]. An online search was conducted on eight databases including SPORTDiscus, CINAHL Complete, MEDLINE (EBSCO), Education Source, Web of Science, Scopus, Embase and Pubmed with no limitation on publication date.

ACCEPTED MANUSCR

The following key words and Boolean terms were searched within major topics, titles and abstracts: 'sport' AND 'ankle' AND ['injury' OR 'sprain'] AND ['etiolog*' OR 'aetiolog*' OR 'correlat*' OR 'risk factor*' OR 'susceptib*' OR 'associat*' OR 'caus*' OR 'contribut*' OR 'predi*'].

Studies were eligible for inclusion if they were published in a peer-reviewed academic journal written in English and retrievable as full-text. For full inclusion and exclusion criteria see Table 1. Upon completion of the search strategy by the lead author (PD), duplicates were removed through EndNote X9.3.1 (version 19.2.0.13018). In order to validate the study selection process, two independent reviewers (PD & IK) screened all records found across the eight databases according to the pre-determined list of inclusion and exclusion criteria. Titles and abstracts were initially read, followed by the full-texts. Reference lists of full texts and systematic reviews were screened further to identify any missed prospective studies.

Table 1 Inclusion and Exclusion Criterion

PICO	Inclusion	Exclusion
Population	Team field-based	Non-field sports
	Contact sport	Non-contact sports
	Male and female	Paralympians
		Enrolled as physical education students
Intervention	Injuries monitored for minimum one competitive season Etiological Studies	Tournaments, showcases, special events Intervention Studies
Comparison	Prospective cohort study	Ankle injuries not defined as sprain
	Player baseline assessments (intrinsic)	Epidemiological studies only
Outcome	Ankle sprain injury incidence Ankle sprain injury risk	Injury rates/risk associations including lower extremity injuries not specific to ankle sprain

2.2 Assessment of Studies and Data Extraction

In order to assess the quality of the studies selected for this review, a modification of the Critical Appraisal Skills Programme-Diagnostic Test Study (CASP) [57] was used with some categories sorted or excluded according to the designs of included studies (i.e. sport injury etiological studies). From the selected studies, the following data were extracted: injury surveillance duration, participant demographics [sex, age, sport and level of competition], exclusion criteria, definition of injury, definition of athletic exposure, type of ankle sprain investigated, injury diagnosis and reporting protocol, baseline physical assessments, and numerous outcome measures if reported (injury occurrence, athletic exposure, injury rate,

ACCEPTED MANUSCRIPT

ankle sprain grades, number of days missed [severity], and risk association). Tables were used to present the data according to baseline assessment category in association with ankle sprain risk. The authors of this review calculated Cohen's *d* [58] if the mean values and standard deviations (±SD) for injured (INJ) and uninjured (UN) groups were reported, and interpreted as follows (small= 0.2, medium= 0.5, large= .8, very large= 1.3) [59, 60]. Otherwise all data are presented as they were reported in the original studies.

3 RESULTS

3.1 Study selection and characteristics

From eight databases, the online search returned 4,012 records, of which 1,291 duplicates were removed. See Figure 1 for PRISMA screening process results. Of the 2,721 remaining, 17 papers [61-77] fit the inclusion criteria and were included for review. The characteristics of each study are described in Table 2. Nine different sports were represented in the seventeen studies which included lacrosse [62, 63, 73], soccer [61-68, 70-74], field hockey [62, 63], football (American) [69, 76], Rugby league [75], Rugby union [75], Australian rules football [75], Gaelic football [77], and hurling [77]. The modified CASP assessment is presented in Supplementary Table 1.

3.2 Injury surveillance

In terms of method of diagnosis, one study confirmed ankle sprains by MRI [75], with eight using assessment by a medical physician [62-64, 66, 68, 72, 74, 77], two using 'medical staff or personnel' [67, 71], one using a 'certified athletic trainer or physician' [69], and five did not acknowledge confirmation or means of diagnosis [61, 65, 70, 73, 76]. In relation to injury recording, nine studies described their protocol for maintaining injury records [61, 64, 65, 67-70, 75, 76] of which six were maintained by the medical staff as part of the original diagnoses. One study appointed undergraduate students [71], and one had the players themselves maintain their own records [77]. Six studies were unclear in reporting these methods [62, 63, 66, 71, 73, 74]. The incidence rate (no. of sprains / exposure) was not described in ten of the studies [62, 65-69, 71, 72, 76, 77]. From the seven studies that did report overall incidence rate, three studies represented data in (#) sprains/1000 hours: 0.6/1000 h [61], 0.36/1000 h [64], 0.47/1000 h [67], 2.16/1000 h [70], and 0.89/1000 h (Rugby union) & 0.46/1000 h (Rugby league) [75]; with one describing in (#) sprains/person-days (p-d), 1.6/1000 p-d (males) & 2.2/1000 p-d (females) [63] and one reported ankle sprains per 10,000 player exposures (12.8 /10,000 AEs) [73].

Among the seventeen studies included, four studies did not differentiate between types of sprain (LAS, MAS, HAS) or mechanism of injury (i.e. contact vs non-contact) [61, 65, 66, 77]; six investigated inversion sprains only [62-64, 68, 69, 71], while three investigated inversion sprains but excluded contact mechanisms [73, 74, 76]. Additionally, one study investigated only syndesmosis sprains [75]; and two studies reported non-contact mechanism only (no differentiation for type of ankle sprain) [67, 69], of which one reported only LAS in their results [67]. Kofotolis et al. [70] reported ankle sprains by individual ligaments rather than the anatomical region, as well as reported contact and non-contact sprains.

ACCEPTED MANUSCRIPT

Four studies investigated the grades of ankle sprains [62, 63, 66, 76], two of which reported the results [62, 76]. Baumhauer et al. [62] reported 12 Grade-I, 3 Grade-II, and 0 Grade-III injuries from a total of 15 inversion sprains in their study. Among the 15 diagnosed non-contact sprains reported by Tyler et al. [76], 11 were Grade-I, 3 were Grade-II, and 1 was a Grade-III sprain. Severity defined by time-loss was represented in three papers [64, 68, 70]. De Ridder et al. [64] reported an average of moderate severity (8-28 days) per sprain, and a mean time-loss of 22.4 (±11.7) days, while Fransz et al. [68] reported 17.7 (±11.9, range 3-54) days for average time-loss, with 32 of the 45 lateral ankle sprains being severe (>7 days missed). Kofotolis et al. [70] found a mean of 7.01 days lost due to ankle sprains in their study.

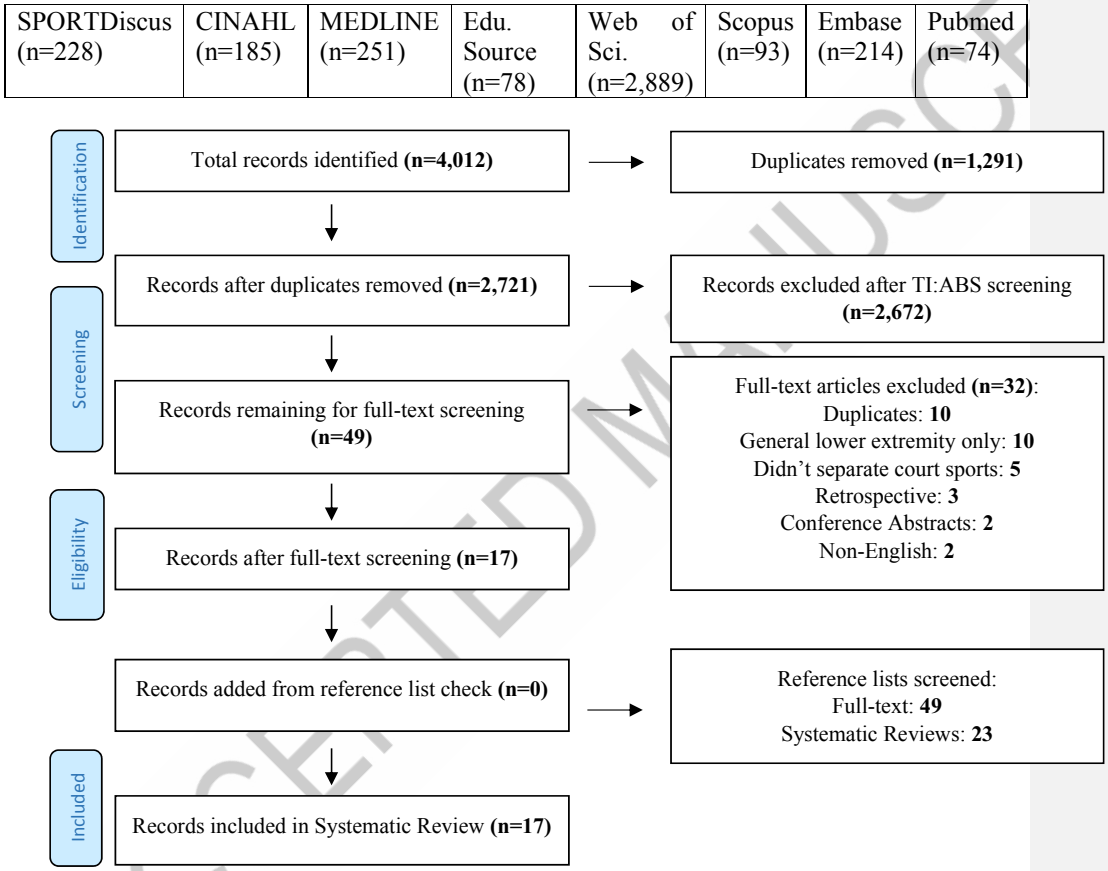

Figure 1: PRISMA flow chart illustrating the selection process for articles in the systematic review

ACCEPTED MANUSCRIPT

Table 2 Description of included studies' characteristics measuring risk factors associated with ankle sprain

Author	Surveillance Period	n= (M, F) Age (y), Sport & Level	Excluded	Type of Sprain & Injury Definition	# Ankle Sprains
Arnason et al. 2004 (Iceland) [61]	1 season	306 (M) 24 y; range 16-38y 18 'best players' across 17 soccer teams in first division	Not reported	Not reported Unable to participate in a match or a training session because of an injury that occurred in a football match or during training	20
Baumhauer et al. 1995 (USA) [62]	1 season	145 (M=73, F=72) Age range 18-23y Intercollegiate lacrosse, soccer or field hockey	Previous grade II or grade III inversion ankle sprain, history of ankle/foot surgery, or regular users of tape/brace	Inversion Inversion ankle sprains were graded I, II, or III based on localized tenderness, swelling, loss of function, and ecchymosis	15 (M= 8, F=7)
Beynnon et al. 2001 (USA) [63]	1 season	118 (M=50, F=68) Age range 18-23y Div I NCAA lacrosse, soccer or field hockey	Previous ankle sprain (questionnaire) Prior ankle/foot surgery (questionnaire)	Inversion If diagnosed, all were graded I, II, or III	20 (M= 7, F=13)
De Ridder et al. 2016 (Belgium) [64]	Tested season 1, assessed injuries for 3 seasons	133 (M) 12.7 y (±2.1); range 10-16y Youth soccer U11-U17	Goalkeepers, injuries from outside soccer, no subjects had current or prior lower extremity injury within 6 mth	Inversion A physical complaint that prohibited a player from participating for at least 48hrs, due to practice or game	12
Ekstrand et al. 1982 (Sweden) [65]	1 season	180 (M) 24.6 y (±4.6) range 17-38y 'Best players' across senior division IV soccer teams	Not reported	Not reported Past injuries were defined as enough for hospital admittance, medical attention, or absence from practice or games for more than a week	63

ACCEPTED MANUSCRIPT

Table 2 (continued) Description of included studies' characteristics measuring risk factors associated with ankle sprain

Author	Surveillance Period	n= (M, F) Age (y), Sport & Level	Excluded	Type of Sprain & Injury Definition	# Ankle Sprains
Ekstrand et al. 1983 (Sweden) [66]	1 year	180 (M) 24.6 y (±4.6) Age range: 17-38y 15 players across 12 senior division soccer teams		Not reported An injury that occurred during a scheduled game or practice and caused a player to miss the next game or practice session	43
Fousekis et al. 2012 (Greece) [67]	1 season	100 (not reported) Professional soccer	Lower extremity injury in previous 6mo	Non-contact Non-contact ankle sprain that forced absence ≥1 scheduled practice/game	17
Fransz et al. 2018 (Amsterdam) [68]	Tested seasons 1&2, assessed injuries for 4 seasons	190 (M) U13, 11.8 y (±.6) U15, 13.9 y (±.6) U17, 15.7 y (±.8) U19, 17.7 y (±.7) 2nd/1st Team, 23.2 y (±3.2) Int-l soccer (AFC Ajax)	Not reported	Inversion Occurring at the lateral ankle and classified as 'sprain' or 'ligamentous injury'	45
Gribble et al. 2015 (USA) [69]	1 season	539 (not reported) 17.33 y (±2.28) HS and Div I NCAA football (American)	Players who suffered lower extremity injuries other than inversion sprains	Inversion 1) Occurred during practice/competition; 2) Required medical attention; 3) ≥1 day of time-loss from football activities; 4) Defined as an incident involving excessive supination; 5) Inability to bear weight; 6) Swelling	54

ACCEPTED MANUSCRIPT

Table 2 (continued) Description of included studies' characteristics measuring risk factors associated with ankle sprain

ACCEPTED MANUSCRIPT

Table 2 (continued) Description of included studies' characteristics measuring risk factors associated with ankle sprain

Author	Surveillance Period	n (M, F) Age (SD) Sport & Level of competition	Excluded	Excluded	Type of Sprain & Injury Definition (Summarized)	# Ankle Sprains
Kofotolis et al. 2007 (Greece) [70]	2 seasons	312 (M) 24.8 y (±4.63) North division soccer league of the Greek Amateur Association	Injuries unrelated to soccer and absence related to illness		By ligament Sustained during training or competition that prevented the injured player from participating in practice or training sessions	139
McCann et al. 2018 (USA) [71]	1 competitive season	43 (F) 19.7 y (±1.1) Div I NCAA soccer	Not reported		Inversion 1) Occurred during team practice/competition; 2) Required care by medical personnel; 3) ≥1 day of missed soccer activity	8
McCunn et al. 2018 (Australia) [72]	1 season	306 (M) 22 y (±4) Semi-professional soccer	Contact injuries		Non-contact All non-contact, time-loss	41
Mineta et al. 2021 (Japan) [73]	16 months	179 (M=76, F=103) 19.8 y Collegiate soccer and lacrosse players	History of lower extremity injury within 3 months or lower extremity surgery. Contact injuries while an athlete was being tackled		Non-contact inversion Injury occurring in connection with practicing sports, with an inversion mechanism that required the player to miss at least one game or practice session.	29

ACCEPTED MANUSCRIPT

	Period	Age (y), Sport & Level		Injury Definition	Sprains
Powers et al. 2017 (Iran) [74]	1 season	185 (M) 20.9 y (±5.9) Age range: 14-34 y 5th Division soccer	Any lower extremity injury in past 6 mo. or history of knee surgery A single ankle sprain (first) per player was counted	Non-contact inversion If non-contact, >1 week absence physical activity	25
Sman et al. 2014 (Australia) [75]	1 season each	202 (M) 21 y (±3.3) Professional rugby league (3 clubs), amateur level rugby union (1 club), Australian football league (1 club)	Not reported	Syndesmosis Occurred during team practice/competition resulting in ≥1 missed/modified practice/game	12
Tyler et al. 2006 (USA) [76]	Two teams 1 season, and two teams 3 seasons	152 (not reported) High School varsity and junior varsity football (American)	Contact sprains occurred while being tackled (force applied by another player)	Non-contact inversion Ankle injury with an inversion mechanism requiring the player to miss ≥1 missed practice/game	15
Watson 1999 (Ireland) [77]	4 years	80 (M) 21.15 y (±3.71) Gaelic football, hurling, or both	Not reported	Did not specify Required medical treatment or that resulted in ≥1 week absence from sport/restricted activity	122

ACCEPTED MANUSCRIPT

3.3 Risk Factors for Ankle Sprain

Across the twelve studies, the following categories were investigated: age, sex, injury history, sport, foot/leg dominance, athletic exposure, anthropometrics, general joint laxity, ankle laxity, anatomical alignment, ankle and hip range of motion, ankle strength, calf strength, center of gravity (balance or similar), muscle reaction time (perturbation), hip strength, movement screens, power, body fat%, respiratory capacity, and flexibility.

Table 3 presents mass, height, BMI and age as risk factors for ankle sprain from studies that reported these findings. In addition to these anthropometric variables, six studies investigated ankle stability by examining ligamentous laxity [61-63, 66, 67, 73]. The Modified Beighton Method (general laxity) was used in two studies showing no differences between the injured and uninjured groups [62, 63]. Joint laxity of the ankle using the anterior drawer test was used in four studies [61-63, 67], where one study assessed with using an ankle arthrometer and an ankle instability questionnaire [73], again all presenting no statistically significant differences between groups. One study reported a general finding (no method reported) of clinical instability of injured players' ankles [66]. Arnason et al. [61], Baumhauer et al. [62], and Beynnon et al. [63] also used a second ankle joint laxity test called the Talar Tilt Test, which only Beynnon et al. [63] found a difference between the injured and uninjured groups for males (p= .002), but not in females. These joint laxity tests were dichotomously scored as positive or negative, with no means nor standard deviations reported, and therefore effect size was not calculated.

ACCEPTED MANUSCRIPT

Table 3 Anthropometrics and Ankle Sprain Risk

Author		UN n=	UN Mean	UN SD	INJ n=	INJ Mean	INJ SD	p-value	Effect Size
De Ridder et al. (LAS) **Male**									**Cohen's d**
	Age (*y)	121	12.5	2.1	12	14.3	1.2	<.001	1.05
	BMI	121	17.6	1.9	12	19.8	1.9	<.001	1.16
Adj. Body Size-	(*BM x HT^2)	121	112	51.5	12	161.3	41.2	.002	1.06
Fousekis et al. (Non-contact)									**Cohen's d**
Male	Age (y)	83	25.4	2.45	17	24.3	5.76		-0.25
	Mass (*kg)	83	71	6.17	17	73.82	5.81		0.47
	Height (*cm)	83	178	5.6	17	177.07	5.93		-0.16
	BMI	83	22.6	0.94	17	23.44	1.42		0.70
	Final Logistic Regression								**OR (CI 95%)**
$^{\beta}$Low Risk	BMI	39			2			0.018	8.16 (1.42, 46.63)
$^{\beta}$High Risk	BMI	44			15				
Low Risk	Mass	51			7			0.017	5.72 (1.37, 23.95)
High Risk	Mass	32			10				
Low Risk	Age	48			13			0.092	.28 (.061, 1.24)
High Risk	Age	35			4				
Gribble et al. (LAS) **Sex not reported**									**Cohen's d**
	Age (y)	485	17.3	2.28	54	17.33	2.28		0.01
	Mass (kg)	485	88	20.8	54	96.15	23.92	0.007	0.36
	Height (cm)	485	181	9.12	54	180.43	8.66		-0.07
	BMI	485	26.7	4.64	54	29.32	6.08	<.001	0.48
	Logistic Regression								**OR (CI 95%)**
	BMI							0.013	2.08 (1.16, 3.72)
Kofotolis et al. (by ligament)	Age (y)								**OR (CI 95%)** 1.09 (0.97, 1.13)
McCann et al. (LAS) **Male**									**Cohen's d**
	Mass (kg)	35	60.6	4.1	8	60.7	6.1	0.96	0.02
	Height (cm)	35	166	3.7	8	169.2	2.3	0.01	1.04
	BMI	35	22	1.5	8	21.2	2.2	0.22	-0.42
	Binary Logistic Regression								**OR (CI 95%)**
	BMI							0.22	0.74 (.46, 1.2)
	Mass							0.95	1.0 (.84, 1.19)
	Height							0.05	1.3 (1.00, 1.70)
	Fisher's Exact Test for Height								$^{\Omega}$**DOR**
	≥167.6 (cm)	17			7				7.5
	<167.6 (cm)	18			1				
Mineta et al. (LAS, Non-contact)									**Cohen's d**
	Age (y)	138	19.8	1.0	29	19.9	1.1	0.529	.100
	Height (cm)	138	165.9	8.2	29	164.7	7.8	0.313	-.146
	Mass (kg)	138	61.0	8.5	29	60.7	8.7	0.599	-.035
Powers et al. (LAS, Non-contact)									**Cohen's d**
	Mass (kg)	160	68.6	10.3	25	71.6	7.1	>0.15	0.34
Male	Height (cm)	160	178	8	25	178	6	>0.15	0.00
	BMI	160	21.6	2	25	22.5	1.1	0.002	0.56

ACCEPTED MANUSCRIPT

UN: uninjured group; **INJ**: injured group

Table 3 (continued) Anthropometrics and Ankle Sprain Risk

Author			UN n=	UN Mean	UN SD	INJ n=	INJ Mean	INJ SD	p-value	Effect Size
Tyler et al. (LAS, Non-contact)	**Sex not reported**									**Incidence Rate (CI 95%)**
α'Normal weight'			61			3				.52 (.1, 1.6)
α 'Risk of overweight'			45			5				1.05 (.3, 2.5)
α 'Overweight'			31			7				2.03 (.8, 4.2)
Sman et al. (HAS)	**Male**									**Cohen's d**
		Age (y)	190	21	3.3	12	20	2.7	0.51	-0.33
		Mass (kg)	190	95.2	13	12	92.5	10	0.48	-0.23
		Height (cm)	190	185	5.9	12	182.1	6.2	0.16	-0.48
		BMI	190	27.9	3.1	12	27.9	2.2	0.98	0.00
Watson (not specified)										**Cohen's d**
		Height (cm)	28	176.3	6.2	52	180.1	6.0	t 2.64	0.62
		BMI	28	23.4	1.5	52	22.42	1.559	t 2.69	-0.64
Baumhauer et al. (LAS)	**Male**									**Cohen's d**
		Mass (kg)	65	79.4	8.5	8	79.0	7.6		-0.05
		Height (cm)	65	178.1	6.35	8	180.3	6.1		0.37
	Female	Mass (kg)	65	61.7	7.8	7	62.5	5.5		0.12
		Height (cm)	65	166.4	7.1	7	165.1	4.3		-0.22
Beynnon et al. (LAS)	**Male**									**Cohen's d**
		Mass (kg)	43	76.7	7.2	7	78.0	4.1		0.23
		Height (cm)	43	178.8		7	183.6			
	Female	Mass (kg)	55	60.3	6.4	13	58.1	6.4		-0.36
		Height (cm)	55	165.4		13	167.1			

*****y= years, BM= body mass; HT= height, kg= kilogram, cm= centimetre
α CDC BMI for Age (Normal weight= 5th-84.9th %; Risk for Overweight= 85th-94.9th %; Overweight= ≥ 95th %)
β Fousekis et al.: Low Risk = below median; High Risk = above median
Ω DOR= diagnostic odds ratio

Anatomical alignment of the ankle was assessed in three studies, with two using goniometric measurements [62, 63] and one using camera (Nikon 35 mm) views anteriorly, laterally, and posteriorly (scoring 'good, moderate deviation, or marked deviation') [77]. Five studies assessed ankle or hip range of motion as a risk factor [61, 65-67, 75], none of which reported associated risk with ankle sprains. The results from four studies which analyzed ankle strength as an associated risk factor are presented in Table 4 [62, 63, 67, 75]. Strength measures were assessed by repetition to fatigue, hand-held dynamometer, and isokinetic dynamometry. Using the Standing Heel Rise Test to assess calf strength, Sman et al. [75] investigated maximal repetitions to fatigue, observing 22.4 (±10.3) repetitions for the injured group compared to 19.9 (±8.2) in the uninjured group (p= .54) (Cohen's d= .305).

ACCEPTED MANUSCRIPT

Table 4 Ankle Strength and Ankle Sprain

Author		UN n=	UN Mean	UN SD	INJ n=	INJ Mean	INJ SD	p-value	Effect Size
Baumhauer et al. (LAS)	**Isokinetic (30º/s) Cybex 6k Dynamometer (Ft-lbs)**								**Cohen's d**
M&F	Plantar flexion $^{\beta}$CON	130	66.24	22.12	15	70.27	21.25	0.504	0.19
	Dorsiflexion CON	130	22.53	5.89	15	24.67	6.28	0.188	0.35
	Inversion CON	130	21.57	7.19	15	22.80	4.93	0.686	0.20
	Eversion CON	130	18.21	4.86	15	20.57	6.38	0.088	0.42
	DF : PF CON	130	0.357	0.074	15	0.360	0.061	0.877	0.04
	Ever : Inv CON	130	0.894	0.193	15	1.00	0.218	0.044	0.52
	Within Injured Group (control= uninjured ankle)								
	Plantar flexion CON	15	68.33	19.26	15	72.20	23.33	0.036	0.18
	Dorsiflexion CON	15	24.93	6.27	15	24.40	6.67	0.518	-0.08
	Inversion CON	15	20.47	4.80	15	21.13	5.69	0.496	0.13
	Eversion CON	15	20.47	6.99	15	20.67	6.32	0.842	0.03
	DF : PF CON	15	0.373	0.068	15	0.348	0.061	0.031	-0.39
	Ever : Inv CON	15	1.00	0.253	15	1.00	0.271	0.987	0.00
Beynnon et al. (LAS)	**Isokinetic (30º/s) Cybex 6k Dynamometer (unit not reported)**								**Cohen's d**
Male	Plantar flexion CON	43	71.80	13.70	7	77.30	13.00	0.340	0.41
	Dorsiflexion CON	43	26.80	3.80	7	28.10	4.22	0.420	0.32
	Inversion CON	43	20.00	5.30	7	21.10	5.11	0.920	0.21
	Eversion CON	43	19.50	5.30	7	22.30	5.66	0.430	0.51
	Plantar flexion $^{\beta}$ECC	43	-89.70	20.90	7	-97.00	32.55	0.310	-0.27
	Dorsiflexion ECC	43	-39.40	6.10	7	-38.90	6.11	0.620	0.08
	Inversion ECC	43	-23.80	6.20	7	-25.70	6.22	0.380	-0.31
	Eversion ECC	43	-24.60	5.80	7	-24.00	5.44	0.850	0.11
	DF/PF CON	43	0.39	0.09	7	0.37	0.09	0.740	-0.22
	Ev/Inv CON	43	1.02	0.30	7	1.07	0.26	0.730	0.18
	DF/PF ECC	43	0.47	0.15	7	0.44	0.17	0.750	-0.19
	Ev/Inv ECC	43	1.08	0.30	7	0.96	0.19	0.210	-0.48
Female	Plantar flexion CON	55	53.10	11.30	13	52.30	12.30	0.960	-0.07
	Dorsiflexion CON	55	18.20	2.60	13	17.00	2.40	0.100	-0.48
	Inversion CON	55	14.20	4.20	13	15.20	5.90	0.750	0.20
	Eversion CON	55	13.30	2.70	13	12.60	3.40	0.470	-0.23
	Plantar flexion ECC	55	-68.40	18.10	13	-68.50	16.30	1.000	-0.01
	Dorsiflexion ECC	55	-26.80	4.10	13	-26.10	4.00	0.340	0.17
	Inversion ECC	55	-16.30	4.30	13	-17.30	3.70	0.840	-0.25
	Eversion ECC	55	-16.20	3.80	13	-15.50	3.30	0.310	0.20
	DF/PF CON	55	0.36	0.09	13	0.34	0.08	0.250	-0.23
	Ev/Inv CON	55	0.99	0.30	13	0.88	0.24	0.330	-0.40
	DF/PF ECC	55	0.42	0.11	13	0.40	0.09	0.490	-0.20
	Ev/Inv ECC	55	1.03	0.24	13	0.92	0.25	0.170	-0.45

ACCEPTED MANUSCR

Table 4 (continued) Ankle Strength and Ankle Sprain

Fousekis et al. (Non-contact)	**Isokinetic (60º/s) BiodexSystem III Dynamometer (N-m)**								**Cohen's d**
	Left Ankle								
	Dorsiflexion CON	83	23.23	5.04	17	22.91	5.00		-0.06
	Dorsiflexion ECC	83	49.86	9.02	17	48.51	10.21		-0.14
	Plantarflexion CON	83	115.60	22.23	17	114.59	21.30		-0.05
	Plantarflexion ECC	83	207.63	36.07	17	212.48	40.68		0.13
	Right Ankle								
	Dorsiflexion CON	83	23.84	5.55	17	21.55	4.23		-0.46
	Dorsiflexion ECC	83	49.72	8.09	17	47.46	7.64		-0.29
	Plantarflexion CON	83	118.05	23.13	17	115.39	23.70		-0.11
	Plantarflexion ECC	83	213.19	39.27	17	224.04	53.67		0.23
	Final Logistic Regression								**OR (95% CI)**
	ECC [α]High Risk	25			9			0.005	8.88 (1.95, 40.36)
	ECC [α]Low Risk	58			8				
Sman et al. (HAS)	**Isometric Lafayette Push Pull Hand-held Dynamometer**								**Cohen's d**
	Ev : Inv (Left)	190	1.09	0.30	12	1.05	0.30	0.610	-0.13
	Ev : Inv (Right)	190	1.01	0.30	12	0.95	0.40	0.410	-0.17

β CON= concentric; ECC= eccentric. **α** High Risk= > 15% asymmetry; Low Risk= < 15% asymmetry.

ACCEPTED MANUSCR

Table 5 Hip Strength and Ankle Sprain

Author		UN n=	UN Mean	UN SD	INJ n=	INJ Mean	INJ SD	p-value	Effect Size
De Ridder et al. (LAS)	**(Isometric) Microfet Hand-Dynamometer (N)**								**Cohen's d**
	Flexors	121	98.10	26.10	12	122.20	41.80	<.001	0.69
(N)/Adj. Body Size		121	2.20	0.70	12	1.70	0.40	0.003	-0.88
	Extensors	121	156.60	34.30	12	207.90	69.60	0.027	0.93
(N)/Adj. Body Size		121	1.60	0.60	12	1.30	0.50	0.070	-0.54
	Abduction	121	145.60	48.30	12	183.50	61.00	0.013	0.69
(N)/Adj. Body Size		121	1.40	0.40	12	1.20	0.50	0.108	-0.44
	Adduction	121	119.20	38.80	12	150.50	64.20	0.014	0.59
(N)/Adj. Body Size		121	1.60	0.60	12	1.30	0.50	0.106	-0.54
	Exorotation	121	100.40	29.20	12	135.10	29.10	<.001	1.19
(N)/Adj. Body Size		121	1.00	0.40	12	0.90	0.20	0.193	-0.32
	Endorotation	121	98.10	26.10	12	122.20	41.80	0.074	0.69
(N)/Adj. Body Size		121	1.00	0.40	12	0.80	0.20	0.066	-0.63
									***HR (CI 95%)**
Body Size Anterior Chain/Horizontal Plane								0.058	1.74 (.980, 3.092)
Posterior Chain								0.028	.33 (.123, .890)
Hip Adduction								0.320	.70 (.344, 1.417)
Hip Abduction								0.463	1.32 (.626, 2.801)
Powers et al. (LAS, Non-contact)	**(Isometric) Commander Power Track II Hand Dynamometer**								**OR (CI 95%)**
	Hip abduction, %BW							0.010	1.1 (1.02, 1.18)

*Cohen's d (effect size): small= 0.2, medium= 0.5, large= .8, very large= 1.3 [57]
**Odds Ratio (effect size): small= 1.5, medium= 2.0, large= 3.0 [57]

Two studies used hand-dynamometers to measure isometric hip strength in relation to ankle sprain risk [64, 74], and both data sets are presented in Table 5. Two studies investigated power as a potential risk factor. Sman et al. [75] used a vertical counter-movement jump with arm swing by way of either a Vertec™ or jump testing mat, where those with ankle sprain (n=12) jumped 63.6 (±8.20) cm during baseline testing compared to those without sprain who achieved 59.10 (±7.8) cm (p= .05) (Cohen's d= .577). Arnason et al. [61] investigated ankle sprain association to a squat test, a countermovement jump, and a standing jump to which no differences between injured and non-injured groups were found. Body fat %, respiratory capacity, and flexibility were investigated by Arnason et al. [61] and Watson et al. [77] but only one stated no significant association to ankle sprain where the other [77] referred to them in their methods but were not reported in the study's results. Risk factors described by authors as center of gravity, balance, proprioception, postural sway or movement screens are compiled to one category (kinematics and kinetics) for representation in Supplementary Table 2 [63, 67-69, 72, 73, 75, 77].

4 DISCUSSION

4.1 Summary of Main Findings

This study is the first review of ankle sprain risk factors measured in a comprehensive list of field-based, team contact sports. Data from the seventeen studies included in the review came

ACCEPTED MANUSCRIPT

from nine different sports including the sports of lacrosse, soccer, field hockey, American football, rugby league, rugby union, Australian rules football, Gaelic football, and hurling. In total, twenty different categories were assessed as potential risk factors, twelve of which showed an association to ankle sprain incidence [63]. Deficits or asymmetries in ankle strength [62, 67] and hip strength [64, 74], and greater BMI [67, 76] were all associated with an increased likelihood of sustaining an ankle sprain in two different studies each, with single leg landing performance outcomes [68, 73] and single leg anterior reach associated in others [69]. Height [71] and mass [67] were independently associated with ankle sprain risk in other cohorts. Finally, joint laxity was associated with ankle sprain incidence in one study [63] while anatomic alignment of the foot and ankle represented as a risk factor in two different studies [62, 63].

4.2 Comparison to Existing Literature

Injury events in sport will involve differing kinematics and kinetics thus affecting different anatomical structures [78]. While ankle sprain prevalence across sports has been established, further epidemiological dissection of specific ankle sprain etiology is needed. Varying grades of ankle sprain have previously been defined to aid different treatment strategies [79]. Indeed, diagnosing sprains by grade is one way to categorize severity; but it would prove more beneficial for coaches and medical staff to additionally communicate the impact through estimated time-loss to relevant stakeholders during the course of a season, rather than the architectural state of injured structures [22]. While injury severity (time-loss) is a calculation made retrospectively at the end of a season, only three studies in this review reported average time-loss per ankle sprain [64, 68, 70]. Furthermore, Bahr et al. have suggested there is a need for sport injury epidemiological studies to report injury burden (combining injury severity and incidence rate) to better manage overall risk in the sporting setting [80].

4.3 Injury Surveillance

For prospectively defining injury, nine studies included time-loss as part of their injury definition, with two studies having required at least a week of absence [74, 77], one required a modified [75] and two required missed [67, 76] game or practice. Two others specified 1 day of missed sport activity [69, 71], one at least 48 hours of missed participation [64], and one noted 'time-loss' with no specified time window [72]. In order to continue evolving from epidemiological studies to etiological studies, practitioners and researchers should standardize records and reports according to consensus statements published for many sports [81-85], including the most recent consensus statement by the International Olympic Committee [86].

4.4 Anthropometric Risk Factors

Anthropometric assessments were the most commonly measured category across the seventeen studies. From the ten studies that measured BMI as an independent risk factor, effect sizes varied from medium to large. While De Ridder et al. [64] reported a higher BMI as a significant risk factor with a large effect size, and Fousekis et al. [67] an 8-fold increase in ankle sprain risk for a BMI above the sample median, Watson et al. [77] on the other hand found a negative relationship suggesting that a higher BMI may be protective against ankle

ACCEPTED MANUSCR

sprains. The findings of Watson et al. [77] should be considered with caution as the type of ankle sprains were not reported in their inclusion nor exclusion criteria, and there was a lack of clarity reported in the methods section of this study as evident in CASP assessment. A higher BMI as indicative of risk, similar to the results of De Ridder et al. [64] and Fousekis et al. [67] has also been found previously for general lower extremity injury [87]. It is worth noting that the methods used to measure BMI differed across studies, that is, three studies were vague in their methods of recording mass and height [64, 76, 77], two used a standardized physician scale [67, 71], one did not acknowledge these methods [74], and four utilized self-reporting [62, 63, 69, 75]. This non-invasive recall method does provide a way of saving practitioners time in team settings, while also showing good reliability in recent studies [88], but it would not be considered the gold standard. An additional body size classification (BW x HT^2) was investigated originally by Milgrom et al. [89], and found positive association for ankle sprain incidence in military recruits. Similarly, De Ridder et al. [64] showed statistical significance and a large effect size when comparing the injured and uninjured groups in their study. Therefore, further studies are warranted to include this adjusted body size in their methods as a potential risk factor for ankle sprains.

4.5 Foot and Ankle Risk Factors

Anatomical alignment of the foot and ankle as measured by a goniometer was associated with increased risk of injury in two studies [62, 63]. Dorsiflexion range of motion was considered as a factor for risk in three studies [67, 75, 77], but results were reported in only one [75]. Sman et al. [75] used the weight-bearing lunge test which has shown to have 'good' inter- and intra-tester reliability, while also showing efficacy for time, space and cost. However, they reported no relationship and a small effect size (Cohen's d= -.121) between the injured group and uninjured group for dorsiflexion, therefore more evidence is needed to include dorsiflexion as a risk factor. Two [62, 67] of the four [62, 63, 67, 75] studies which measured ankle strength as a risk factor found significant differences between groups. Baumhauer et al. [62] found that between groups, the ratio of concentric eversion:inversion strength was higher in the injured group, and showed a medium effect size (Cohen's d= 0.52). The same study reported bilateral values within the injured player, finding statistically significant differences but small effect sizes between the injured and uninjured limb in concentric plantarflexion (Cohen's d= 0.18) and the ratio of concentric dorsiflexion:plantarflexion (Cohen's d= -.39). Fousekis at al. [67] reported an 8.88 odds ratio for those with a 15% or greater asymmetry in eccentric dorsiflexion and eccentric plantarflexion strength. In an investigation of 241 male physical education students, Willems et al. [90] showed that decreased concentric dorsiflexion strength relative to body mass was also associated with inversion sprains. When ankles are at higher risk due to lack of intrinsic strength or stability [91], many practitioners may argue for the use of external support in prevention of injury for or when returning from injury [92]. However, the only study included in this review to also assess external support as a risk factor [76], showed a significantly higher incidence of 3.9/1,000 athlete-exposures (95% CI, 1.8-7.4) for non-contact inversion sprains for those using tape, braces or both compared to those that did not use any external support 0.5/1,000 athlete-exposures (95% CI, 0.2-1.1) ($p < .001$). Furthermore, a recent meta-analysis has also shown that for patients with history of ankle sprain, external support showed no significant improvement in dynamic

ACCEPTED MANUSCRIPT

postural control for those who did not use support [93], indicating further the importance of decreasing risk of injury onset.

4.6 Lower Body Kinematic and Kinetic Risk Factors

Three of the eight studies that investigated balance-like traits found significant relationships of moderate to large effect for those who incurred an ankle sprain [68, 69, 73]. Using a force plate, Fransz et al. [68] assessed single leg depth drop landing performance (20 cm height), showing a combined risk factor with stability in the medio-lateral direction within the first 0.4 seconds of landing, and stability within the horizontal direction between seconds 3.0 and 5.0. As seen in Supplementary Table 2, other independent variables (i.e. peak force vertical, root mean square medio-lateral, mean horizontal late dynamic phase) associated with severe ankle sprains (>1-week time-loss). The second study in this category to show a difference between groups was Gribble et al. [69] using the SEBT (anterior reach) (OR= 2.84), while Sman et al. [75] found no difference in risk of sprain using the SEBT. In 2021 across 167 college soccer and lacrosse athletes, Mineta et al. used an eight infra-red camera motion analysis system and forceplate to measure kinetic and kinematic data during single leg landing tasks. They found that when knee varus angle is greater than 0.2º of knee valgus angle there is more than a five-fold increase risk of non-contact LAS, and 24 times greater when pelvic internal rotation is greater than 6º [73]. Neither of the two studies which investigated a modified FMS [69] nor Soccer Injury Movement Screen [72] showed an association with ankle sprains . These mixed results among movement screen scores agree with previously published studies suggesting contradictory findings for general injury association [94, 95]. The inconsistencies across performance tests and methods of statistical analysis throughout this review are particularly evident within this category, inhibiting a collective interpretation across data.

4.7 Hip Strength

Hip external rotation strength has been considered and demonstrated as a key characteristic for correct landing particularly for mechanics of the knee [96, 97]. De Ridder et al. [64] utilized a hand-held dynamometer to show that decreased posterior hip strength was associated with a greater risk of sustaining a lateral ankle sprain. The second study [74] that tested hip strength found a value of 33.8% of body mass in hip abduction strength was the cut-off for increased risk of sustaining injury. That is to say, for every 1% decrease in relative strength in hip abduction, the risk of injury increases by 10% [74]. When considering the kinetic chain, this evidence warrants further investigation into the role of hip strength and distal ankle sprain injury.

4.8 Injury History

While this paper reviewed prospective risk factors, many of the included studies also collected questionnaire data regarding athletes' previous injury, and only two did not [72, 77]. Two in fact used previous injury as an exclusion criterion [64, 73]. Baumhauer et al. [62] reported that during their investigation, 27% of those who sustained an ankle sprain had a previous Grade I injury in the same ankle, where Gribble et al. [69] reported 35% had prior ankle sprain history, 75% were recurrent sprains in McCann et al. [71], and Powers et al. [74]

ACCEPTED MANUSCR

reported 32% of injured had a history of ankle sprain. In addition to reporting previous injuries relative to acquired sprains during the study duration, Sman et al. [75] performed a Backward Stepwise Cox Regression to find as part of a univariate and multivariate analysis that previous sprain was not statistically significant for risk. Contradictory to that, Tyler et al. [76] reported previous ankle sprain as an independent risk factor for future sprain, and when combining with being overweight (BMI, >95% for age) it resulted in an incidence of 4.27 (95% CI 1.6, 9.3) compared to 0.22 (95% CI .01, 1.2) in normal weighted persons with no previous injury. Similar to history of ankle sprain as a risk factor, Mineta et al. [73] reported no group difference in the Identification of Functional Ankle Instability questionnaire scores between the LAS and non-injured groups. While these historical measures were all determined through recall, it does identify a risk factor which should be more reliably measured in future studies. For example, a longitudinal injury surveillance study would have injury records for each player over the course of several competitive seasons and calendar years, allowing this risk factor to be measured prospectively with less bias.

Strengths and Limitations

The current review performed a comprehensive search of eight databases and included prospective studies which assessed baseline measurements before a competitive season in field-based team contact sports. By narrowing the search to this population during the screening process, confounding variables were better controlled for (e.g. sporting rules, footwear, and surface). Additionally, by utilizing studies with a prospective cohort design, studies with recall bias were eliminated. There was significant heterogeneity described across the methods and for the included studies and this inhibited pooled quantitative analyses, although a recent meta-analysis has taken this approach [51]. Only papers in English were included in this review. Out of the 17 included studies, only 4 studied women. Other limitations to this review include not setting an age range for subjects used for inclusion, as connective tissue (e.g. ligament) maturation may differ between adolescents and adults. However, if a meta-synthesis was to be performed, they could be excluded in the statistical analysis, as the distal tibial epiphysis closes between 12 and 15 years of age [98].

Implications for Practice and Research

From a practical standpoint, even though different sports may have high rates of ankle sprain, as etiology may vary, coaches need to be able to develop specific programs for their working population (i.e. needs analysis). For example, if a strength and conditioning coach is responsible for physical development of a university's women's lacrosse and men's basketball team, this may be complicated by the fact that these sports result in different types of sprains, that is, lateral ankle sprains and high ankle sprains, respectively. Without evidence of associated risk factors for each type of sprain independently, the injury reduction plan for ankle sprains will likely be a one-model-fits-all approach which may not be appropriate.

Soccer was the most frequently investigated sport throughout this review, represented in thirteen of the seventeen studies. Females were only represented in four of the studies. This suggests a lack of evidence for prospective studies measuring ankle sprain risk factors in field-based team contact sports and across both sexes. It is worth acknowledging that during

ACCEPTED MANUSCRIPT

the screening process, ten papers were excluded for investigating risk factors for general lower extremity injuries rather than reporting specific location and type (i.e. ankle sprain). A further five papers did not separate their findings by sport (i.e. population), and the current authors could not obtain these data from those corresponding authors, thus those papers were excluded from this review. To date, epidemiological studies have identified the specific ligaments involved in knee sprains, yet continue to ignore separating types of ankle sprain (LAS, MAS, HAS) in their reports [32], ignoring the different structures, kinetics, and mechanisms involved. To progress the impact of applied research, all studies and papers investigating injury risk factors should adhere to the newly published IOC guidelines for injury recording and reporting [86]. This standard of practice will aid quantitative conclusions from level I and level II research [99], identifying which risk factors have highest sensitivity and specificity for injury risk screening [100].

When assessing player baseline risk factors, validity and reliability of chosen assessments need to be considered, with time and cost effectiveness also playing an important role. Furthermore, all studies included herein used one single assessment period during preseason. In some cases baseline testing was only performed once while participants were surveyed across multiple seasons without retesting. In order to increase the validity of the findings, more research needs to be completed to describe normative changes in these risk factors over the course of competitive season(s) and determine how often an assessment should be implemented. Accurate documentation for player attendance in training sessions and matches will provide more accurate exposure-time and injury incidence rates. When publishing research, extensive and detailed appendices should be attached for reviewers to access describing means (±SD) for injured and uninjured groups, confidence intervals, as well as providing data separated by sex, sport type, and playing position. As previously suggested, when consensus statements are not able to be followed, studies should describe which resources were limited in their design that prevented them from doing so [85].

Many challenges to conducting prospective etiological research in sports injuries have been laid out previously by Bahr [100]. His paper identifies appropriate and successive steps to validate a screening test, yet indicates that no such test has been appropriately justified to predict sporting injuries. This support, along with the important quantitative limitations from this review into ankle sprain risk factors, reinforce that practitioners should design interventions around injury incidence frequencies rather than potential risk of causative factors. Furthermore, specific programs which target multiple body parts should be considered, as sports elicit many injury types across various bodily locations.

5 CONCLUSIONS

This review has two overall findings, firstly from more than twenty categories of risk factors investigated for ankle sprain association across seventeen studies in field-based team contact sports, 12 variables were found to be associated with increased incidence of ankle sprain in field-based team contact sport athletes. From the studies reviewed, no movement screen scores including from a modified-FMS nor Soccer Injury Movement Screen showed an increased risk of sustaining ankle sprain. In order to reduce the risk of ankle sprain, practical

ACCEPTED MANUSCRIPT

assessment of BMI, ankle plantar and dorsiflexion strength, hip strength, and single leg landing performance should be factored in to athlete assessment and subsequent program design. Ankle sprain interventions must be included in a whole-body program in order to maximize the transfer from research to pragmatically reducing injuries. In addition to these practitioner focused elements, considering that ankle sprains are the most prevalent injury across all sports, from this review it remains clear that more prospective studies with large sample sizes across both sexes, utilizing standardized definitions and methods of recording and reporting are needed. While etiological studies will identify risk factors, there will always be an inherent risk of injury in sport. Therefore, applied research will intend to reduce injury risk, occurrence or severity, rather than prevent.

Declarations

Ethics All procedures set out in this paper have been approved by the relevant institution's ethics committee

Disclosure of interest All authors declare no conflict of interest that are relevant to the content of this review.

Availability of Data The original datasets analyzed from included studies during the current review are available from the corresponding author upon request.

References

1. WHO, *The World Health Report, Reducing Risks, Promoting Healthy Life_2002.pdf.* 2002.
2. McMahon, E.M., et al., *Physical activity in European adolescents and associations with anxiety, depression and well-being.* Eur Child Adolesc Psychiatry, 2017. **26**(1): p. 111-122.
3. Sun, F., I.J. Norman, and A.E. While, *Physical activity in older people: a systematic review.* BMC public health, 2013. **13**(1): p. 449.
4. Prakash, R.S., et al., *Physical activity and cognitive vitality.* Annu Rev Psychol, 2015. **66**: p. 769-97.
5. Harsha, D.W. and G.S. Berenson, *The benefits of physical activity in childhood.* The American journal of the medical sciences, 1995. **310**: p. S109-S113.
6. Bauman, A., et al., *The International Prevalence Study on Physical Activity: results from 20 countries.* International Journal of Behavioral Nutrition and Physical Activity, 2009. **6**(1): p. 21.
7. Maher, C.A., M. Toohey, and M. Ferguson, *Physical activity predicts quality of life and happiness in children and adolescents with cerebral palsy.* Disabil Rehabil, 2016. **38**(9): p. 865-9.
8. Duman, S., H.D. Sukan, and M. Ceylan, *The Impact of Physical Activity on Socializing Mentally Handicapped Children.* Internationa Journal of Humanities and Social Science Invention, 2016. **5**(3): p. 49-51.
9. Poyatos-Leon, R., et al., *Effects of exercise-based interventions on postpartum depression: A meta-analysis of randomized controlled trials.* Birth, 2017. **44**(3): p. 200-208.
10. Watson, A., et al., *Associations between organised sport participation and classroom behaviour outcomes among primary school-aged children.* PLoS One, 2019. **14**(1): p. e0209354.

ACCEPTED MANUSCRIPT

11. Andersen, M.H., L. Ottesen, and L.F. Thing, *The social and psychological health outcomes of team sport participation in adults: An integrative review of research.* Scand J Public Health, 2019. **47**(8): p. 832-850.
12. Kokkinos, P., *Physical activity, health benefits, and mortality risk.* ISRN cardiology, 2012. **2012**.
13. Gajawelli, N., et al., *Neuroimaging changes in the brain in contact versus noncontact sport athletes using diffusion tensor imaging.* World neurosurgery, 2013. **80**(6): p. 824-828.
14. Kemler, E., et al., *Ankle injuries in the Netherlands: Trends over 10-25 years.* Scand J Med Sci Sports, 2015. **25**(3): p. 331-7.
15. Boyce, S.H. and M.A. Quigley, *Review of sports injuries presenting to an accident and emergency department.* Emerg Med J, 2004. **21**(6): p. 704-6.
16. Lambers, K., D. Ootes, and D. Ring, *Incidence of patients with lower extremity injuries presenting to US emergency departments by anatomic region, disease category, and age.* Clin Orthop Relat Res, 2012. **470**(1): p. 284-90.
17. Rae, K. and J. Orchard, *The Orchard sports injury classification system (OSICS) version 10.* Clinical Journal of Sport Medicine, 2007. **17**(3): p. 201-204.
18. Waterman, B.R., et al., *The epidemiology of ankle sprains in the United States.* J Bone Joint Surg Am, 2010. **92**(13): p. 2279-84.
19. Hootman, J.M., R. Dick, and J. Agel, *Epidemiology of Collegiate Injuries for 15 Sports: Summary and Recommendations for Injury Prevention Initiatives.* Journal of Athletic Training (National Athletic Trainers' Association), 2007. **42**(2): p. 311-319.
20. Fong, D.T.-P., et al., *A systematic review on ankle injury and ankle sprain in sports.* Sports medicine, 2007. **37**(1): p. 73-94.
21. Aoki, H., et al., *A 15-year prospective epidemiological account of acute traumatic injuries during official professional soccer league matches in Japan.* Am J Sports Med, 2012. **40**(5): p. 1006-14.
22. Whalan, M., et al., *The incidence and burden of time loss injury in Australian men's sub-elite football (soccer): A single season prospective cohort study.* Journal of science and medicine in sport, 2019. **22**(1): p. 42-47.
23. Knowles, S.B., et al., *Cost of injuries from a prospective cohort study of North Carolina high school athletes.* Inj Prev, 2007. **13**(6): p. 416-21.
24. Hupperets, M.D.W., E.A.L.M. Verhagen, and W. van Mechelen, *Effect of unsupervised home based proprioceptive training on recurrences of ankle sprain: randomised controlled trial.* BMJ (Clinical Research Ed.), 2009. **339**: p. b2684-b2684.
25. Hansen, J.T., *Netter's Clinical Anatomy E-Book*. 2017: Elsevier Health Sciences.
26. Swenson, D.M., et al., *Epidemiology of US High School Sports-Related Ligamentous Ankle Injuries, 2005/06-2010/11.* Clinical Journal of Sport Medicine, 2013. **23**(3): p. 190-196.
27. Heiss, R., et al., *Prevalence of MRI-detected ankle injuries in athletes in the Rio de Janeiro 2016 Summer Olympics.* Academic radiology, 2019. **26**(12): p. 1605-1617.
28. Eric Chehab, M.D. *HSS: The High Ankle Sprain: Whats' the Difference*. [web] 2019; Available from: https://www.hss.edu/conditions_high-ankle-sprain-whats-different.asp.
29. Sports, M.U., *Ankle Sprain Symptoms, Diagnosis, and Treatment | UPMC.* 2020.
30. Enright, K., et al., *Workload and Injury in Professional Soccer Players: Role of Injury Tissue Type and Injury Severity.* Int J Sports Med, 2020. **41**(2): p. 89-97.
31. Edouard, P., et al., *Injury frequency and characteristics (location, type, cause and severity) differed significantly among athletics ('track and field') disciplines during 14 international championships (2007-2018): implications for medical service planning.* Br J Sports Med, 2020. **54**(3): p. 159-167.
32. Moreno-Alcaraz, V.J., A. Cejudo, and P. Sainz de Baranda, *Injury types and frequency in Spanish inline hockey players.* Phys Ther Sport, 2020. **42**: p. 91-99.

ACCEPTED MANUSCRIPT

33. Ogaki, R., et al., *Incidence, Severity, and Burden of Ankle Sprains in Male Collegiate Rugby Union Players.* International Journal of Sport and Health Science, 2022: p. 202117.
34. Bielska, I.A., et al., *The health economics of ankle and foot sprains and fractures: A systematic review of English-language published papers. Part 1: Overview and critical appraisal.* Foot (Edinb), 2019. **39**: p. 106-114.
35. Mailuhu, A.K.E., et al., *Clinical and radiological predictors for persistent complaints five years after a lateral ankle sprain: A long-term follow-up study in primary care.* Journal of Science and Medicine in Sport, 2018. **21**(3): p. 250-256.
36. Swenson, D.M., et al., *Patterns of recurrent injuries among US high school athletes, 2005-2008.* Am J Sports Med, 2009. **37**(8): p. 1586-93.
37. Van Mechelen, W., H. Hlobil, and H.C. Kemper, *Incidence, severity, aetiology and prevention of sports injuries.* Sports medicine, 1992. **14**(2): p. 82-99.
38. Finch, C., *A new framework for research leading to sports injury prevention.* J Sci Med Sport, 2006. **9**(1-2): p. 3-9; discussion 10.
39. Taimela, S., U.M. Kujala, and K. Osterman, *Intrinsic Risk Factors and Athletic Injuries.* Sports Medicine, 1990. **9**(4): p. 205-215.
40. Theisen, D., et al., *Injuries in Youth Sports: Epidemiology, Risk Factors and Prevention.* Deutsche Zeitschrift für Sportmedizin, 2014. **2014**(09): p. 248-248.
41. Orchard, J.W. and J.W. Powell, *Risk of Knee and Ankle Sprains under Various Weather Conditions in American Football.* Medicine & Science in Sports & Exercise, 2003. **35**(7): p. 1118-1123.
42. Bishop, D., *The effects of travel on team performance in the Australian national netball competition.* Journal of science and medicine in sport, 2004. **7**(1): p. 118-122.
43. Rice, S.G., M. American Academy of Pediatrics Council on Sports, and Fitness, *Medical conditions affecting sports participation.* Pediatrics, 2008. **121**(4): p. 841-8.
44. Green, T., et al., *What is the quality of clinical practice guidelines for the treatment of acute lateral ankle ligament sprains in adults? A systematic review.* BMC Musculoskelet Disord, 2019. **20**(1): p. 394.
45. Huang, F., et al., *Acupuncture for the treatment of ankle sprain: A protocol for a systematic review and meta-analysis: study protocol.* Medicine (Baltimore), 2019. **98**(46): p. e17905.
46. Petersen, W., et al., *Treatment of acute ankle ligament injuries: a systematic review.* Arch Orthop Trauma Surg, 2013. **133**(8): p. 1129-41.
47. Wikstrom, E.A., C. Mueller, and M.S. Cain, *Lack of Consensus on Return to Sport Criteria Following Lateral Ankle Sprain: A Systematic Review of Expert Opinions.* J Sport Rehabil, 2019: p. 1-20.
48. Doherty, C., et al., *Treatment and prevention of acute and recurrent ankle sprain: an overview of systematic reviews with meta-analysis.* Br J Sports Med, 2017. **51**(2): p. 113-125.
49. Pourkazemi, F., et al., *Predictors of chronic ankle instability after an index lateral ankle sprain: A systematic review.* Journal of Science & Medicine in Sport, 2014. **17**(6): p. 568-573.
50. Beynnon, B.D., D.F. Murphy, and D.M. Alosa, *Predictive factors for lateral ankle sprains: a literature review.* Journal of athletic training, 2002. **37**(4): p. 376.
51. Kobayashi, T., M. Tanaka, and M. Shida, *Intrinsic Risk Factors of Lateral Ankle Sprain: A Systematic Review and Meta-analysis.* Sports Health: A Multidisciplinary Approach, 2016. **8**(2): p. 190-193.
52. Correia, M., *Intrinsic and Extrinsic Risk Factors for Lateral Ankle Sprain: A Literature Review.* Archives of Sports Medicine, 2019. **3**(2).
53. de Noronha, M., et al., *Do voluntary strength, proprioception, range of motion, or postural sway predict occurrence of lateral ankle sprain?* British Journal Of Sports Medicine, 2006. **40**(10): p. 824-828.

ACCEPTED MANUSCRIPT

54. Witchalls, J., et al., *Intrinsic functional deficits associated with increased risk of ankle injuries: a systematic review with meta-analysis.* British Journal of Sports Medicine, 2012. **46**(7): p. 515-523.
55. Ryan, J., N. DeBurca, and K. Mc Creesh, *Risk factors for groin/hip injuries in field-based sports: a systematic review.* Br J Sports Med, 2014. **48**(14): p. 1089-96.
56. Shamseer, L., et al., *Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation.* BMJ : British Medical Journal, 2015. **349**: p. g7647.
57. Critical Appraisal Skills Programme, C., *CASP Checklist-Diagnostic Test Study*. 2018.
58. Ellis, P.D., *Effect Size Calculators.* https://www.polyu.edu.hk/mm/effectsizefaqs/calculator/calculator.html, 2020.
59. Sullivan, G.M. and R. Feinn, *Using Effect Size-or Why the P Value Is Not Enough.* J Grad Med Educ, 2012. **4**(3): p. 279-82.
60. Cohen, J., *Statistical power analysis for the behavioral sciences*. 2013: Routledge.
61. Arnason, A., et al., *Risk Factors for Injuries in Football.* American Journal of Sports Medicine, 2004. **32**(1): p. 5S-16S.
62. Baumhauer, J.F., et al., *A Prospective Study of Ankle Injury Risk Factors.* The American Journal of Sports Medicine, 1995. **23**(5): p. 564-570.
63. Beynnon, B.D., et al., *Ankle ligament injury risk factors: a prospective study of college athletes.* Journal Of Orthopaedic Research: Official Publication Of The Orthopaedic Research Society, 2001. **19**(2): p. 213-220.
64. De Ridder, R., et al., *Hip Strength as an Intrinsic Risk Factor for Lateral Ankle Sprains in Youth Soccer Players: A 3-Season Prospective Study.* Am J Sports Med, 2017. **45**(2): p. 410-416.
65. Ekstrand, J. and J. Gillquist, *The frequency of muscle tightness and injuries in soccer players.* The American Journal Of Sports Medicine, 1982. **10**(2): p. 75-78.
66. Ekstrand, J. and J. Gillquist, *The avoidability of soccer injuries.* International journal of sports medicine, 1983. **4**(02): p. 124-128.
67. Fousekis, K., E. Tsepis, and G. Vagenas, *Intrinsic risk factors of noncontact ankle sprains in soccer: a prospective study on 100 professional players.* Am J Sports Med, 2012. **40**(8): p. 1842-50.
68. Fransz, D.P., et al., *Performance on a Single-Legged Drop-Jump Landing Test Is Related to Increased Risk of Lateral Ankle Sprains Among Male Elite Soccer Players: A 3-Year Prospective Cohort Study.* Am J Sports Med, 2018. **46**(14): p. 3454-3462.
69. Gribble, P.A.p.g.u.e., et al., *Prediction of Lateral Ankle Sprains in Football Players Based on Clinical Tests and Body Mass Index.* American Journal of Sports Medicine, 2016. **44**(2): p. 460-467.
70. Kofotolis, N.D., E. Kellis, and S.P. Vlachopoulos, *Erratum: Ankle Sprain Injuries and Risk Factors in Amateur Soccer Players During a 2-Year Period.* American Journal of Sports Medicine, 2007. **35**(5): p. 849-849.
71. McCann, R.S., et al., *Acute lateral ankle sprain prediction in collegiate women's soccer players.* International Journal of Sports Physical Therapy, 2018. **13**(1): p. 12-18.
72. McCunn, R., et al., *Soccer Injury Movement Screen (SIMS) Composite Score Is Not Associated With Injury Among Semiprofessional Soccer Players.* Journal of Orthopaedic & Sports Physical Therapy, 2018. **48**(8): p. 630-+.
73. Mineta, S., et al., *Greater knee varus angle and pelvic internal rotation after landing are predictive factors of a non-contact lateral ankle sprain.* Physical Therapy in Sport, 2021. **50**: p. 59-64.
74. Powers, C.M.p.u.e., et al., *Hip Strength as a Predictor of Ankle Sprains in Male Soccer Players: A Prospective Study.* Journal of Athletic Training (Allen Press), 2017. **52**(11): p. 1048-1055.
75. Sman, A.D., et al., *Predictive factors for ankle syndesmosis injury in football players: A prospective study.* Journal of Science & Medicine in Sport, 2014. **17**(6): p. 586-590.

ACCEPTED MANUSCRIPT

76. Tyler, T.F., et al., *Risk Factors for Noncontact Ankle Sprains in High School Football Players: The Role of Previous Ankle Sprains and Body Mass Index.* American Journal of Sports Medicine, 2006. **34**(3): p. 471-475.
77. Watson, A.W.S., *Ankle sprains in players of the field-games Gaelic football and hurling.* Journal of Sports Medicine & Physical Fitness, 1999. **39**(1): p. 66-70.
78. Fallat, L., D.J. Grimm, and J.A. Saracco, *Sprained ankle syndrome: prevalence and analysis of 639 acute injuries.* The Journal of foot and ankle surgery, 1998. **37**(4): p. 280-285.
79. Cass, J.R. and B.F. Morrey, *Ankle Instability: Current Concepts, Diagnosis, and Treatment.* Mayo Clinic Proceedings, 1984. **59**(3): p. 165-170.
80. Bahr, R., B. Clarsen, and J. Ekstrand, *Why we should focus on the burden of injuries and illnesses, not just their incidence*. 2018, BMJ Publishing Group Ltd and British Association of Sport and Exercise Medicine.
81. Fuller, C.W., et al., *Consensus statement on injury definitions and data collection procedures in studies of football (soccer) injuries.* Scand J Med Sci Sports, 2006. **16**(2): p. 83-92.
82. Fuller, C.W., et al., *Consensus statement on injury definitions and data collection procedures for studies of injuries in rugby union.* Br J Sports Med, 2007. **41**(5): p. 328-31.
83. Mountjoy, M., et al., *Consensus statement on the methodology of injury and illness surveillance in FINA (aquatic sports).* Br J Sports Med, 2016. **50**(10): p. 590-6.
84. Orchard, J.W., et al., *International consensus statement on injury surveillance in cricket: a 2016 update.* Br J Sports Med, 2016. **50**(20): p. 1245-1251.
85. Brown, J.C., et al., *Guidelines for community-based injury surveillance in rugby union.* J Sci Med Sport, 2019. **22**(12): p. 1314-1318.
86. Bahr, R., et al., *International Olympic Committee Consensus Statement: Methods for Recording and Reporting of Epidemiological Data on Injury and Illness in Sports 2020 (Including the STROBE Extension for Sports Injury and Illness Surveillance (STROBE-SIIS)).* Orthopaedic Journal of Sports Medicine, 2020. **8**(2).
87. Nilstad, A., et al., *Risk factors for lower extremity injuries in elite female soccer players.* Am J Sports Med, 2014. **42**(4): p. 940-8.
88. Davies, A., et al., *Validity of self-reported weight and height for BMI classification: A cross-sectional study among young adults.* Nutrition, 2020. **71**: p. 110622.
89. Milgrom, C., et al., *Risk Factors for Lateral Ankle Sprain: A Prospective Study Among Military Recruits.* Foot & Ankle International, 1991. **12**(1): p. 26-30.
90. Willems, T.M., et al., *Intrinsic risk factors for inversion ankle sprains in females – a prospective study.* Scandinavian Journal of Medicine & Science in Sports, 2005. **15**(5): p. 336-345.
91. Kaminski, T.W., et al., *National Athletic Trainers' Association Position Statement: Conservative Management and Prevention of Ankle Sprains in Athletes.* Journal of Athletic Training (Allen Press), 2013. **48**(4): p. 528-545.
92. DiSipio, A., et al., *What is the most effective external ankle support to speed recovery in patients with ankle sprain?* Evidence-Based Practice, 2020. **23**(2): p. 19-20.
93. Tsikopoulos, K., et al., *Do External Supports Improve Dynamic Balance in Patients with Chronic Ankle Instability? A Network Meta-analysis.* A Publication of The Association of Bone and Joint Surgeons® | CORR®, 2020. **478**(2): p. 359-377.
94. Chalmers, S., et al., *Asymmetry during preseason Functional Movement Screen testing is associated with injury during a junior Australian football season.* Journal of science and medicine in sport, 2017. **20**(7): p. 653-657.
95. Newton, F., et al., *Functional Movement Screen (FMS™) score does not predict injury in English Premier League youth academy football players.* Science and Medicine in Football, 2017. **1**(2): p. 102-106.
96. Lawrence III, R.K., et al., *Influences of hip external rotation strength on knee mechanics during single-leg drop landings in females.* Clinical biomechanics, 2008. **23**(6): p. 806-813.

ACCEPTED MANUSCRIPT

97. Malloy, P., et al., *Hip external rotator strength is associated with better dynamic control of the lower extremity during landing tasks.* Journal of strength and conditioning research, 2016. **30**(1): p. 282.
98. Rosenbaum, A.J., J.A. DiPreta, and R.L. Uhl, *Review of distal tibial epiphyseal transitional fractures.* Orthopedics, 2012. **35**(12): p. 1046-9.
99. Burns, P.B., R.J. Rohrich, and K.C. Chung, *The levels of evidence and their role in evidence-based medicine.* Plastic and reconstructive surgery, 2011. **128**(1): p. 305.
100. Bahr, R., *Why screening tests to predict injury do not work-and probably never will...: a critical review.* Br J Sports Med, 2016. **50**(13): p. 776-80.

ACCEPTED MANUSCR

Supplementary Table 1

	Was there a clear question for the study to address?	Did all patients get the diagnostic test and reference standard?	Could the results of the test have been influenced by the results of the reference standard?	Is the disease status of the tested population clearly described?	Was there a comparison with an appropriate reference standard? **Body Composition**	Were the methods for performing the test described in sufficient detail?	Was there a comparison with an appropriate reference standard? **Ankle Stability**	Were the methods for performing the test described in sufficient detail?	Was there a comparison with an appropriate reference standard? **Knee Stability**	Were the methods for performing the test described in sufficient detail?	Was there a comparison with an appropriate reference standard? **Ankle Strength**	Were the methods for performing the test described in sufficient detail?	Was there a comparison with an appropriate reference standard? **Knee Strength**	Were the methods for performing the test described in sufficient detail?
Arnason [61]	**Y**	**Y**	**N**	**Y**	**Y**	**Y**	**Y**	**Y**	**Y**	**Y**				
Baumhauer [60]	**Y**	**Y**	**N**	**U**			**Y**	**Y**			**Y**	**Y**		
Beynnon [61]	**Y**	**Y**	**N**	**U**			**Y**	**Y**			**U**	**Y**		
De Ridder [62]	**Y**	**Y**	**N**	**Y**	**Y**	**U**								
Ekstrand [63]	**Y**	**Y**	**U**	**U**					**Y**	**U**				
Ekstrand [64]	**Y**	**Y**	**N**	**Y**			**Y**	**Y**					**Y**	**Y**
Fousekis [65]	**Y**	**Y**	**N**	**Y**	**Y**	**Y**	**Y**	**Y**			**U**	**U**		
Fransz [66]	**Y**	**Y**	**N**	**U**										
Gribble [67]	**Y**	**Y**	**U**	**Y**	**Y**	**U**								
Kofotolis [68]	**Y**	**Y**	**N**	**Y**	**Y**	**Y**								
McCann [69]	**Y**	**Y**	**N**	**Y**	**Y**	**Y**								
McCunn [70]	**Y**	**Y**	**N**	**U**										
Mineta [71]	**Y**	**Y**	**N**	**Y**			**Y**	**Y**						
Powers [72]	**Y**	**Y**	**N**	**Y**	**Y**	**N**								
Sman [73]	**Y**	**Y**	**U**	**U**	**Y**	**Y**					**U**	**U**		
Tyler [74]	**Y**	**Y**	**N**	**Y**	**Y**	**U**								
Watson [75]	**Y**	**Y**	**U**	**U**	**Y**	**U**								

Y: yes; N: no; U: unclear; blank: study did not investigate corresponding risk factor.

ACCEPTED MANUSCRIPT

Supplementary Table 1 (continued)

	Was there a comparison with an appropriate reference standard? **Ankle/Foot Alignment**	Were methods for performing the test described in sufficient detail?	Was the comparison with an appropriate standard? **Lower Body Kinematics and Kinetics**	Were the methods for the test described in sufficient detail?	Was the comparison with an appropriate standard? **Muscle Reaction Time**	Were methods for the test described in sufficient detail?	Was the comparison with an appropriate reference standard? **Hip Strength**	Were the methods for the test described in sufficient detail?	Was the comparison with an appropriate standard? **Range of Motion**	Were methods for the test described in sufficient detail?	Was the comparison with an appropriate standard? **Calf Muscle Fatigue**	Were the methods for the test described in sufficient detail?	Was the comparison with an appropriate reference standard? **Power**	Were the methods for the test described in sufficient detail?	Was the comparison with an appropriate reference standard? **Clinical Deficiencies**	Were the methods for the test described in sufficient detail?	Was the comparison with an appropriate reference standard? **Respiratory Capacity**	Were the methods for the test described in sufficient detail?
Arnason									**Y**	**Y**			**Y**	**Y**			**Y**	**Y**
Baumhauer	**Y**	**Y**																
Beynnon	**Y**	**Y**	**U**	**U**	**U**	**U**												
De Ridder							**Y**	**Y**										
Ekstrand									**Y**	**U**								
Ekstrand									**Y**	**U**								
Fousekis			**U**	**N**					**Y**	**U**								
Fransz			**U**	**U**														
Gribble			**Y**	**Y**														
Kofotolis																		
McCann																		
McCunn			**Y**	**Y**														
Mineta			**Y**	**Y**														
Powers							**U**	**U**										
Sman	**Y**	**Y**	**U**	**U**					**Y**	**U**	**Y**	**Y**	**Y**	**U**				
Tyler																		
Watson	**U**	**N**	**N**	**U**											**N**	**N**	**U**	**N**

Y: yes; N: no; U: unclear; blank: study did not investigate corresponding risk factor.

ACCEPTED MANUSCRIPT

Supplementary Table 2 Kinematics and Kinetics and Ankle Sprain

Author	UN n=	UN Mean	UN SD	INJ n=	INJ Mean	INJ SD	p-value	Effect Size
Beynnon et al. [61] (LAS)	**Neurotest Force Plate - Anterior-Posterior COG (degrees of sway)**							**Cohen's d**
(M) Single leg stance non-sway	43	3.00	0.24	7	3.60	0.55	0.210	1.41
(M) Single leg stance sway	43	4.80	0.77	7	4.90	1.10	0.590	0.11
(M) Two leg stance sway	43	3.80	0.39	7	3.40	0.36	0.270	-1.07
(F) Single leg stance non-sway	55	3.40	0.33	13	3.40	0.32	0.650	0.00
(F) Single leg stance sway	55	4.80	0.79	13	5.00	0.71	0.250	0.27
(F) Two leg stance sway	55	3.90	0.40	13	4.00	0.64	0.900	0.19
Mineta et al. [71] (LAS, Non-contact)	**Infra-red camera Oqus Motion Analysis & Kistler Force Plate**							**Cohen's d**
Knee varus angle at IC	138	-1.6	3.1	29	2.1	4.1	<.001	1.1
Pelvic anterior tilt peak angle	138	2	3.0	29	1.3	2.1		.23
Pelvic obliquity peak angle	138	2.7	5.0	29	5.4	3.7		.57
Pelvic internal rotation peak angle	138	2.4	3.2	29	6.3	5.5	<.001	1.03
Hip flexion peak angle	138	43.9	11.3	29	44.8	11.7		.08
Hip abduction peak angle	138	1.2	7.1	29	4.9	6		.53
Hip external rotation peak angle	138	8.3	7.4	29	6.3	10.5		.24
Knee flexion peak angle	138	54.5	9.4	29	53.4	9.9		.11
Knee varus peak angle	138	-3.7	4	29	3.9	5.8	<.001	1.75
Knee external rotation peak angle	138	-3.8	4.4	29	-4.8	4.8		.23
Ankle dorsflexion peak angle	138	26.8	7	29	26.4	10		.05
Ankle inversion peak angle	138	7.4	5.4	29	10	5.7		.47
Ankle external rotation peak angle	138	-.9	6.2	29	1.4	5.2		.37
Knee varus moment	138	-.76	.6	29	.04	.65	<.001	1.31
Fransz et al. [66] (LAS)	**AMTI Force Plate**							***Exp(B)**
Peak Force Vertical							0.149	0.80 (.056, 1.08)
Peak Force Anterior-Posterior							0.767	0.95 (.70, 1.31)
Root Mean Square Medio-lateral							0.017	0.70 (.50, .93)
Mean Horizontal (dynamic phase: .4-2.4s)							0.747	1.05 (.78, 1.42)
Mean Horizontal (late dynamic phase: 3.0-5.0s)							0.072	1.30 (.98, 1.72)
Time To Stabilization VRAW							0.763	0.95 (.71, 1.29)
Multivariate								
Root Mean Square Medio-lateral							0.007	0.66 (.48, .89)
Mean Horizontal (late dynamic phase: 3.0-5.0s)							0.029	1.38 (1.03, 1.83)
Severe Sprains (n=32 >7days)								
Peak Force Vertical							0.026	0.68 (.48, .96)
Peak Force Anterior-Posterior							0.182	0.78 (.53, 1.13)
Root Mean Square Medio-lateral							0.012	0.61 (.42, .90)
Mean Horizontal (dynamic phase: .4-2.4s)							0.362	1.18 (.82, 1.70)
Mean Horizontal (late dynamic phase: 3.0-5.0s)							0.016	1.49 (1.08, 2.05)
Time To Stabilization VRAW							0.426	1.17 (.8, 1.71)
Multivariate								
Root Mean Square Medio-lateral							0.005	0.60 (.41, .86)
Mean Horizontal (late dynamic phase: 3.0-5.0s)							0.006	1.57 (1.14, 2.18)

ACCEPTED MANUSCRIPT

Supplementary Table 2 (continued) Kinematics and Kinetics and Ankle Sprain

Author		UN n=	UN Mean	UN SD	INJ n=	INJ Mean	INJ SD	p-value	Effect Size
Watson [75] (not specified)	**Single leg balance (15s)**								
	Normal >15s	22			16				
	Abnormal <15s	8			31				
Fousekis et al. [65] (Non-contact)	**Proprioception – Neuromuscular Coordination of the Ankle Joint**								
	Kinesthetic Stabilometer Prokin-200								**Cohen's d**
	Deterioration (Left) (%)	83	32.81	17.89	17	30.36	11.15		-0.16
	Deterioration (Right) (%)	83	31.84	13.67	17	28.57	5.76		-0.31
Sman et al. [73] (HAS)	Triple Hop Balance Test - 3 meter minimum with balance								**Cohen's d**
	Left (Pass)	124			7				
	Left (Fail)	66			5			0.630	
	Right (Pass)	130			8				
	Right (Fail)	60			4			0.900	
	Star Excursion Balance Test (SEBT)								
	[α]Composite score (cm)	190	77.90	6.10	12	80.50	5.20	0.140	0.46
Gribble et al. [67] (LAS)	**Star Excursion Balance Test (SEBT)**								**Cohen's d**
	Anterior (% leg length)	485	69.72	7.53	54	65.57	7.90	< .001	-0.54
	Posteriomedial (% leg length)	485	85.76	11.03	54	84.35	11.91	0.422	-0.12
	Posteriolateral (% leg length)	485	74.60	11.38	54	71.29	11.04	0.038	-0.30
	[α]Composite score (cm)	485	77.09	8.45	54	74.14	8.19	0.039	-0.35
	Logistic Regression Analysis								**OR (CI 95%)**
	SEBT-Anterior							< .001	2.84 (1.58, 5.10)
	Modified-Functional Movement Screen								
	Modified-Total Score	485	8.37	2.31	54	7.88	2.58	0.322	-0.200
McCunn et al. [70] (Non-contact)	**Soccer Injury Movement Screen (SIMS)**								**[µ]RR (CI 90%)**
	Composite Score								1.04 (.99, 1.09)
	Anterior reach (objective)								1.06 (.94, 1.20)
	Single leg hop for distance (objective)								1.11 (1.00, 1.23)
	Single leg deadlift (subjective)								1.10 (.95, 1.28)
	In-line lunge (subjective)								.90 (.73, 1.11)
	Tuck jump (subjective)								.97 (.83, 1.14)

*Exp(B) is the ratio of the hazards between 2 players whose outcome value on the risk factor differs by 1 unit (z score) when all other covariates are held constant.

α SEBT Composite Score: mean of anterior, posteriomedial, and posteriolateral

*SIMS Composite Score (total #/50, 0-10 for each subtest); [µ] Relative Risk

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCR